

Asymmetric Synthesis of 3-Substituted Proline Chimeras Bearing Polar Side Chains of Proteinogenic Amino Acids

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The amino–zinc–ene–enolate cyclization reaction is a straightforward route to the synthesis of 3-substituted prolines. Herein we report the application of this reaction to the syntheses of proline chimeras of lysine, glutamic acid, glutamine, arginine, and serine. All these compounds were obtained in enantiomerically pure form and suitably protected for peptide synthesis.

The control of the tridimensional structure of peptides and proteins by chemical modification of proteinogenic amino acids has been the keystone of peptide research for the past few decades. The aim of these studies is the development of compounds with improved selectivity, bioavailability, stability, and permeability.¹ The major difficulty remains in inducing the correct folding of the peptide while retaining the critical recognition elements involved in the interaction.² Among the various approaches (disulfide bridging, lactam cyclization, *N*-methylation, ...) described in the *de novo* design of peptides with a high propensity to fold with predetermined secondary structure, the replacement of a native residue by proline chimeras has been widely used.³ These cyclic amino acids constrain the peptide backbone Φ value around -60° , while insertion of the side chain of natural amino acids on the pyrrolidine ring might give data on both its conformation and the importance of the information it carries. With this aim, we have recently demonstrated that, if the insertion of proline in a heterochiral peptide sequence remains the easiest strategy to induce a β -turn,⁴ incorporation of 3-substituted prolines is a valuable approach to mimic natural β -turn type I, II, and II' found in proteins with retention of the side chain functionality in the $i + 1$ position of the turn.⁵ These secondary structures are known to play several roles such

as stabilizing the tertiary structure, initiating folding, and facilitating intermolecular recognition.⁶ Recently, polyproline II (PPII) helices have received much attention. They are believed to be the dominant conformation of proline-rich sequences⁷ and are often involved in mediating protein–protein interactions.⁸ The introduction of chemical diversity into polyproline II helices through proline chimeras may lead to compounds able to inhibit protein–protein interactions by creating new stabilizing interactions as suggested by Chmieliewski.⁹

Several methods for the synthesis of 3-substituted prolines have been reported, but there is still a need for a general stereoselective route to these compounds.¹⁰ We have already reported the amino–zinc–ene–enolate cyclization as a versatile strategy for the asymmetric synthesis of *cis*-3-alkylprolines.¹¹ We report here the extension of this methodology for the synthesis of proline chimeras bearing polar side chains of natural amino acids and suitably protected for peptide synthesis.

Results and Discussion

The asymmetric syntheses of *cis*-3-prolinolysine, *cis*-3-prolinoglutamine, *cis*-3-prolinoglutamic acid, *cis*-3-prolinoarginine, and *cis*-3-prolinohomoserine were investigated through functionalization of the zinc intermediate **2** obtained after the amino–zinc–enolate cyclization of

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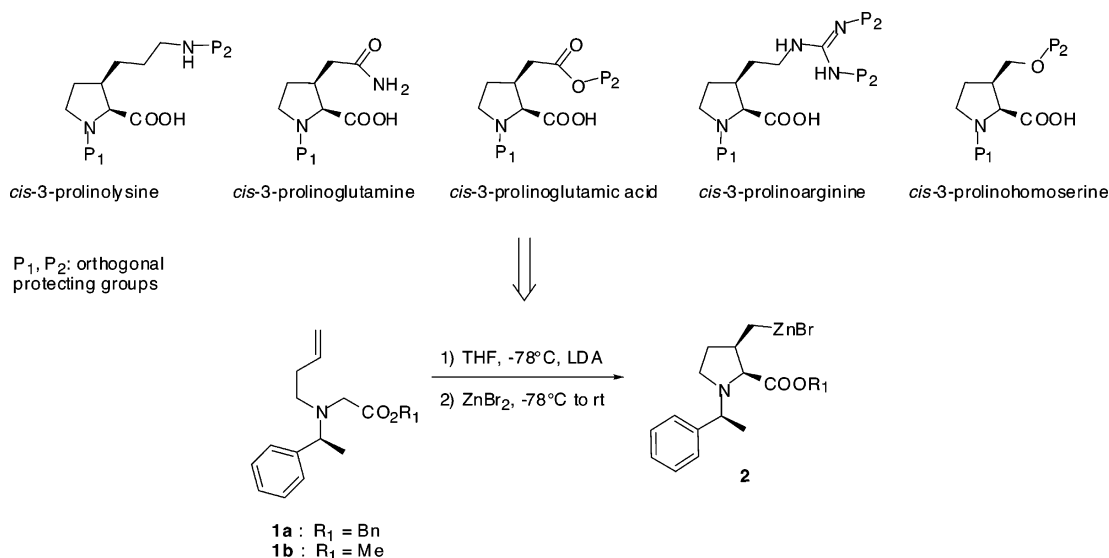
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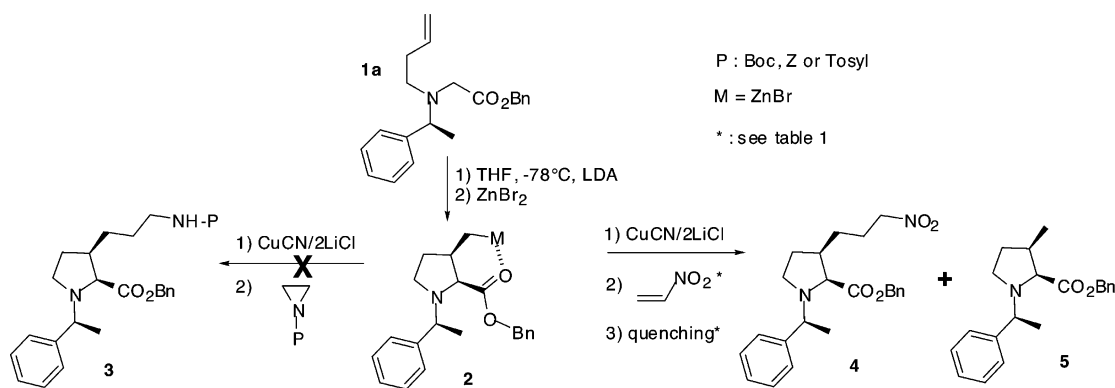
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SCHEME 1



SCHEME 2



olefin **1** (Scheme 1). In a typical procedure, olefin **1** (methyl ester or benzyl ester) was treated with LDA in THF at -78°C , and the lithium enolate was transmetalated with zinc bromide. Warming the reaction mixture to room temperature led to organozinc derivative **2**, with a *cis* relative stereochemistry and 2*S* absolute configuration as previously demonstrated.¹¹

Synthesis of Protected *cis*-3-Prolinolysine **8.** To our knowledge, the synthesis of the titled compound has never been reported. The introduction of the lysine side chain on the cyclic organozinc species **2** was first investigated by aziridine ring opening (Scheme 2). The main interest of this approach was that an ϵ -protected form of 3-prolinolysine would be prepared in a “one-pot” procedure starting from olefin **1a**. Indeed, nonsubstituted aziridine¹² can be considered as an “aminoethyl” synthon, and the aziridine ring opening by organometallic species (organocuprate obtained from lithium or Grignard reagents) has been reviewed.¹³ Moreover, the protection of the aziridine nitrogen by electron-withdrawing groups

(Boc, Z, or tosyl), stabilizing the charge formed after ring opening, has been described as a way to activate this synthon.¹³

The cyclic organozinc reagent **2** is poorly reactive¹⁴ compared to classical organozinc reagents. A chelation of the metal by the ester function after cyclization can explain this low reactivity (Scheme 1). Nevertheless, transmetalation of organozinc compounds by THF-soluble copper salts leads to copper–zinc reagents that have been reported to exhibit reactivity similar to that of lithium or Grignard reagent derived copper organometallics.¹⁵ In the case of the amino–zinc–ene–enolate cyclization leading to **2**, this enhanced reactivity has been demonstrated by the synthesis of a few proline chimeras.¹⁶ Despite that, the opening of *N*-protected aziridines (Boc, Z, or tosyl) by the mixed copper–zinc species was never observed, even upon activation with $\text{BF}_3 \cdot \text{Et}_2\text{O}$. Compound **5**, resulting from hydrolysis of the copper–zinc reagent upon quenching, was the sole product of the reaction.

Confronted with the low reactivity of the organometallic species, we searched for a more reactive aminoethyl

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SCHEME 3

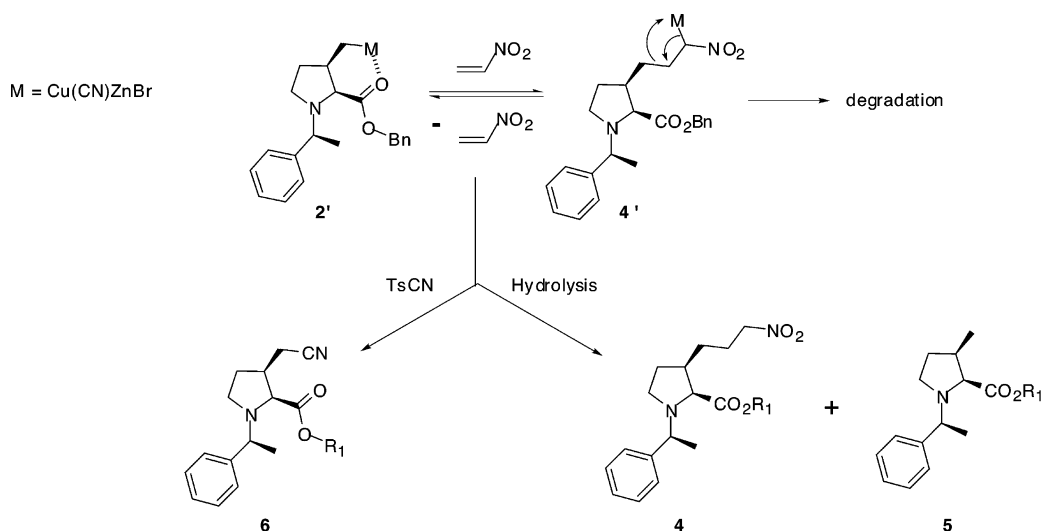


TABLE 1. Copper–Zinc Addition to Nitroethylene

entry	amt of nitroethylene (equiv)	reaction time	reaction temp (°C)	isolated yield (%) of 4 and 5
1	1	15 min	−78	4, 30; 5, 40
2	1	1 h	−78	4, 26; 5, 35
3	1	2 h	−78	4, 23; 5, 33
4	2 at one time	15 min	−78	4, 26; 5, 32
5	3 at one time	15 min	−78	4, 20; 5, 20
6	3 at 15 intervals	1 h	−78	4, 18; 5, 22
7	1	15 min	−100	4, 27; 5, 39
8	1	15 min	−40	4, 25; 5, ND ^a
9	1	15 min	0	4, 2; 5, ND ^a
10	1	15 min	−78 and warming to 0	4, 25; 5, 33

^a ND = not determined.

synthon. Nitroolefins are known to be very good Michael acceptors, and their use for the generation of nitroalkane is well documented.¹⁷ More interestingly, copper–zinc reagents have been shown to add cleanly to nitroolefins.^{17b} Hence, the organozinc species **2** was submitted to transmetalation by the THF-soluble CuCN/2LiCl salts and reacted with nitroethylene under various conditions, some of which are reported in Table 1.

Surprisingly, whatever the reaction conditions, the yield of the Michael adduct **4** never exceeded 30%. The only other major product isolated from the reaction mixture was compound **5**, resulting from hydrolysis of the copper–zinc intermediate. The best reaction conditions (entry 1) were reached with 1 equiv of nitroolefin added at −78 °C followed by 15 min of stirring before quenching at this temperature. Indeed, a longer reaction time (entries 2, 3, and 6), addition of more than 1 equiv of nitroethylene (entries 4–6), or performing the reaction at a temperature higher than −78 °C (entries 8–10) led to lower yields of both the Michael adduct **4** and the

hydrolysis product **5**. However, almost the same **4/5** ratio was observed in each case, just as if an equilibrium between the two organometallic species **4'** and **2'** was governing this reaction (Scheme 3).

It is known that highly stabilized carbanions undergo reversible addition to α,β -unsaturated ketones. Reversible conjugate addition of organocuprates to enones has also been reported.¹⁸ However, reversibility in the addition of copper–zinc reagents to nitroolefins has never been observed.¹⁷ In our case, the reversibility of the Michael addition reaction can be considered if one admits that the organometallic species **2'** is stabilized by chelation between the metal and the ester function (Scheme 2).

To test the reversibility of the carbon–carbon bond forming step, the reaction mixture obtained after addition of the nitroolefin was treated with tosyl cyanide (TsCN). We have already reported that copper–zinc reagent **2'** reacts in good yield with this electrophile.^{16c} Therefore, 15 min after the addition of nitroethylene (corresponding to the conditions of entry 1), 1 equiv of TsCN was added to the mixture, and the solution was stirred at −78 °C for another 2 h (Scheme 2). After quenching and purification, less than 2% of nitro adduct **4** (instead of 30%, entry 1) could be isolated along with 39% of the cyano derivative **6** and 14% of the hydrolysis product **5**. No product resulting from the addition of Michael adduct **4'** to tosyl cyanide was detected. These results confirm the reversibility of the Michael addition.

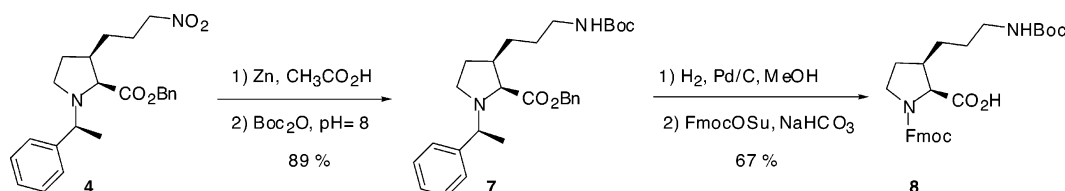
However, the reversibility of the addition is not enough to account for the low yield observed when several equivalents of nitroethylene were added to displace the equilibrium toward the nitro adduct **4**. In this case, degradation and/or polymerization of the enolate resulting from the initial 1,4-addition have to be considered.¹⁹ Indeed, nucleophiles that contain an electron-withdrawing group such as nitrile or nitro groups are among the best nucleophiles in Michael addition. In our case, the Michael addition of the copper–zinc reagent to nitro-

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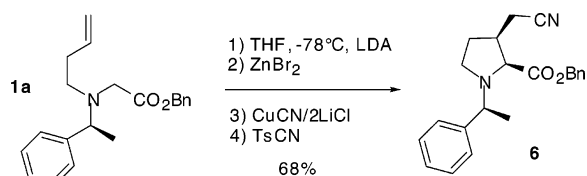
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SCHEME 4



SCHEME 5



ethylene generates a very reactive carbanion (**4'**), which polymerizes and leads to degradation. This point was exemplified (i) by a longer reaction time, which led to a decreasing yield of both compounds **4** and **5** (entries 2 and 3), and (ii) by the use of an excess of nitroolefin (entries 4–6): in that case, in addition to the decreasing yield, a dinitroethyl adduct was detected by ESI⁺ MS analysis in the crude reaction mixture (*m/z* 470.15 [MH⁺]). The use of (TMS)Cl, reported to increase the yield of the Michael addition by trapping the intermediate anion to avoid side reactions,¹⁹ had no effect in our case (data not shown).

Despite the moderate yield (30%), this methodology allows the one-pot preparation of compound **4**, which is a very good intermediate to prepare an orthogonally protected 3-prolinolysine suitable for peptide synthesis (Scheme 4). First, the nitro group was reduced to a primary amine by treatment of **4** with zinc in acetic acid. This primary amine was Boc-protected after filtration on a Celite pad and alkalization, leading to compound **7** in good yield. Special care should be taken to keep the pH ≤ 8; otherwise, Boc₂-amine is obtained as a side product. *N*ε-Boc-*N*α-Fmoc-3-prolinolysine (**8**) was then obtained after catalytic hydrogenation of **7** over palladium charcoal, followed by Fmoc protection.

Synthesis of 3-Prolinoglutamic acid, 3-Prolinoglutamine, and 3-Prolinoarginine. The syntheses of the titled compounds were investigated starting from a common intermediate, i.e., the cyano derivative **6** (Scheme 5), which can be obtained in good yield in a one-pot procedure starting from olefin **1a**.^{16c}

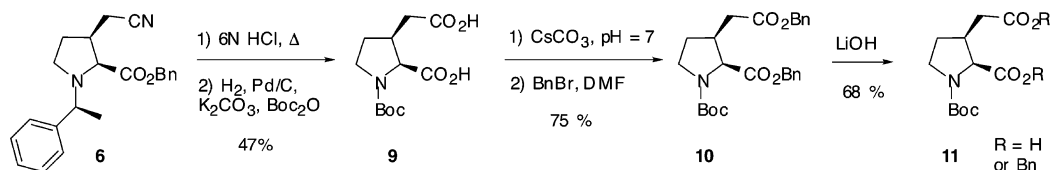
(a) 3-Prolinoglutamic Acid. We have previously reported the synthesis of unprotected *cis*- and *trans*-3-prolinoglutamic acid from **6**.^{16c} In the present study, we have investigated different routes for the preparation of orthogonally protected 3-prolinoglutamic acid, suitable for peptide synthesis. Several methods are described to obtain γ -esters of glutamic acid. The most frequently used is the alkylation of alkali-metal salts of glutamate copper-

(II) complex.²⁰ However, this method is quite tedious and is hardly usable for low gram or milligram scales. Interestingly, it has been shown that saponification of *N*-(benzyloxycarbonyl)aspartic acid dibenzyl ester²¹ and *N*-(benzyloxycarbonyl)glutamic acid bis-2,4,6-trimethylbenzyl ester²² with LiOH occurs selectively on the α -ester, providing the desired β - or γ -ester. This simple method was applied to **6**, via the dibenzyl ester **10** intermediate (Scheme 6).

Hydrolysis of nitrile **6** followed by one-pot hydrogenolysis and Boc protection afforded **9** in moderate yield. Alkylation of the cesium salt of **9** with benzyl bromide gave the desired dibenzyl ester **10**. Saponification of **10** with 1 equiv of LiOH provided one major product (**11**) containing a single benzyl ester. To determine which of the α - or γ -acid group was still protected as a benzyl ester, an HMBC (heteronuclear multiple-bond connectivity) 500 MHz NMR experiment was performed on the isolated major product.²³ A portion of the NMR spectrum is shown in Figure 1A. The correlation of the carbon at 171 ppm with the benzylic protons at 5.18 ppm clearly identifies this carbon as the benzyl ester carbonyl. Correlations are also observed between this carbon and H α (²*J*, 4.45 ppm) and H β (³*J*, 2.82 ppm), indicating that this carbon belongs to the α -carbonyl group. Conversely, the carbon at 176 ppm correlates with the methylene of the C3-substituent. Thus, compound **11** is protected on the α -position. This inverse selectivity in the deprotection reaction probably reflects the difficulties encountered upon the saponification of the α -ester function of *cis*-3-substituted prolines, which often requires heating (see below).

Thus, we searched for conditions which could preserve the benzyl ester of **6** and convert the nitrile into an appropriate ester. The most simple strategy was the use of acetyl chloride in dry MeOH (Scheme 7).²⁴ These mild acidic conditions led in good yield to the γ -methyl ester α -benzyl ester **12**. One-pot hydrogenolysis and Boc protection afforded 3-prolinoglutamic acid γ -methyl ester (**13**), which is suitable for peptide synthesis. Although the position of the ester was not ambiguous, we performed the same HMBC experiment for comparison with α -ester **11** (Figure 1B). The correlation of the carbon at 172 ppm with the methyl protons at 3.70 ppm identifies this carbon as the carbonyl of the methyl ester. This carbon shows one more correlation at 2.45 ppm which is attributed to the methylene of the C3-substituent. The

SCHEME 6



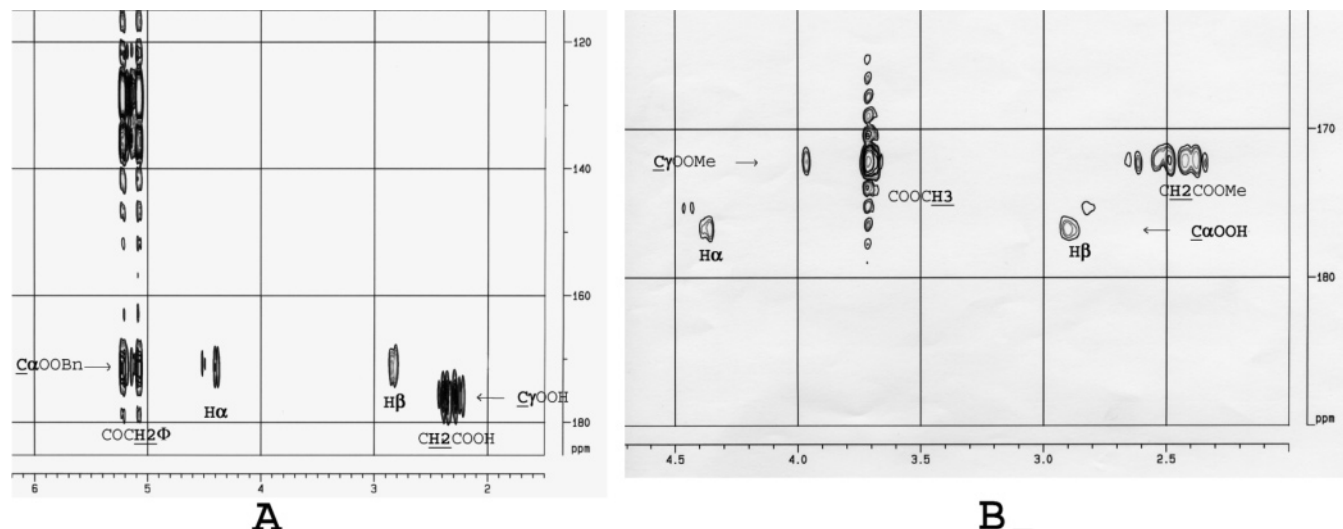
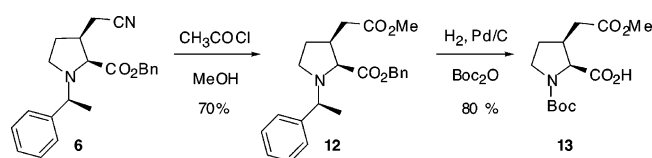


FIGURE 1. Portion of the HMBC spectra of compounds **11** (A) and **13** (B).

SCHEME 7



other carbonyl at 176 ppm correlates with H α (2J , 4.39 ppm) and H β (3J , 2.90 ppm). These results confirmed the γ -position of the methyl ester.

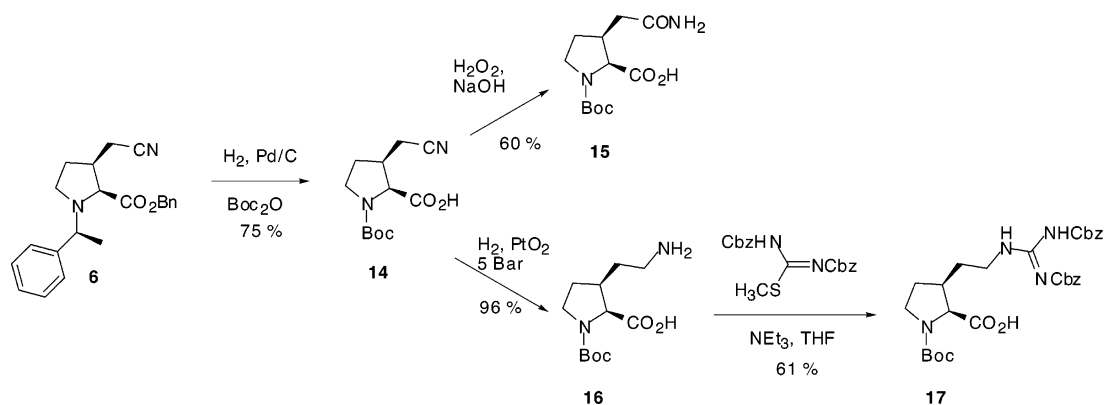
It is worth mentioning that the syntheses of both α - and γ -orthogonally protected 3-prolinoglutamic acids are achievable from the intermediate **6**.

(b) 3-Prolinoglutamine and 3-Prolinoarginine. As previously reported, one-pot debenzoylation and Boc protection of the pyrrolidine ring occur cleanly on such compounds when Boc₂O is added to the hydrogenation mixture.¹¹ Under these conditions (Scheme 8) only the amine and the ester functions of **6** were deprotected, leaving intact the nitrile function. The cyano intermediate **14**, obtained in good yield, turned out to be a versatile compound for the preparation of two chimeras, i.e., prolinoglutamine and prolinoarginine. *cis*-Boc-3-prolino-glutamine (**15**) was first obtained by treatment of **14** with H₂O₂ and NaOH.

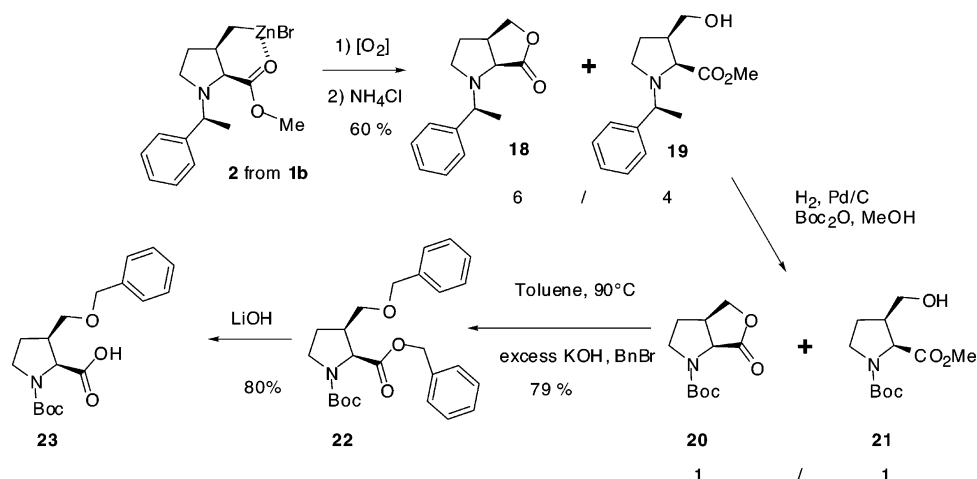
Further reduction of the nitrile function of compound **14** was studied to access prolinoarginine. Several meth-

ods are described for the conversion of nitrile to amine. Complex metal hydrides such as NaBH₄ + CoCl₂·6H₂O have been reported to successfully achieve nitrile–amine conversion,²⁵ and these conditions can be applied to Boc-containing compounds.²⁶ Indeed, reduction of **14** with this method consumed the starting material in a few hours. However, no pure compound could be isolated, probably due to formation of metal chelates and/or decomposition of the products. We therefore focused on reduction of the nitrile to amine by hydrogenation although catalytic hydrogenation of nitriles often gives poor yields. The difficulty seems to arise because the reduction proceeds stepwise via the aldimine, some of which can condense with the primary amine already formed, giving secondary amines.²⁷ Many studies describe the use of Raney nickel catalysis in a basic medium under quite drastic pressure and/or temperature conditions.²⁸ Smoother conditions have been described with the use of a cocatalyst such as Pd/C²⁹ or rhodium.³⁰ A very attractive method is the catalytic hydrogenation of nitriles over PtO₂, in the presence of CHCl₃, which allows the in situ formation of the amine hydrochloride, thus avoiding any side reactions.³¹ In our case, CHCl₃, which could have damaged the Boc protecting group (by in situ HCl formation), turned out to be needless. Indeed, the presence of the carboxylic acid function of compound **14** was sufficient to allow a clean reduction of the nitrile group of **14** over

SCHEME 8



SCHEME 9



PtO₂ in MeOH, under 5 bar of H₂, to afford amine **16** with an excellent yield (96%). Guanilation of **16** using *N,N'*-di-Cbz-*S*-methylisothiurea³² in THF provided orthogonally protected *cis*-3-prolinoarginine **17**.

Synthesis of *cis*-3-Prolinohomoserine. The synthesis of this compound was investigated through direct oxidation of zinc intermediate **2** (Scheme 9). Oxidation of organometallic compounds to alcohols has been previously studied by bubbling dry air³³ or dioxygen³⁴ in THF. More recently, the oxidation of piperidinic organozinc compounds has been studied.^{11d} These procedures have been applied to the organozinc **2** for the synthesis of prolinohomoserine. Direct oxidation of organozinc **2** with dry air did not take place, but bubbling O₂ in the mixture quickly afforded a mixture of lactone **18** and methyl ester **19**. As lactonization occurred until protection of the hydroxyl group, the synthesis was followed up on the mixture. Hydrogenolysis of the mixture over Pd/C and Boc protection in methanol afforded a mixture of lactone **20** and methyl ester **21**. The mixture was then heated at 80 °C in toluene with an excess of KOH and benzylbromide to give compound **22**. Saponification of the benzyl ester required heating overnight with 5 equiv of LiOH. Epimerization (5%) of the α -center was observed by NMR analysis of the crude product. The two diastereoisomers were easily separated by flash chromatography, and *N*-Boc-prolinohomoserine benzyl ether (**23**) was obtained in 80% yield.

Conclusion

Proline chimeras are valuable tools for a wide field of biological applications. We have reported the amino–zinc–enolate cyclization as a straightforward method for the asymmetric synthesis of 3-substituted prolines, and we have extended this methodology to the synthesis of all types of proline chimeras bearing a side chain of natural amino acids. The strategy described in this study allows the asymmetric synthesis, in few steps, of polar 3-substituted prolines suitably protected for peptide synthesis. The introduction of these proline chimeras into biologically active peptides is under way.

Experimental Section

General Considerations. Tetrahydrofuran (THF) and diethyl ether (Et₂O) were distilled from sodium benzophenone and kept over 4 Å molecular sieves. Dry dimethylformamide (DMF) and other reagents were commercially available. Zinc bromide (ZnBr₂) was dried by fusion under flame and nitrogen; the fused salts were allowed to solidify under nitrogen and then dissolved in dry Et₂O. Lithium chloride (LiCl) was dried with a heat gun under reduced pressure. After the LiCl was cooled under nitrogen, CuCN was added, followed by dry THF, and the mixture was then stirred until complete dissolution (1 h). Olefins **1a** and **1b** were prepared as described previously.^{11a} Nitroethylene was prepared from nitroethanol and resublimed phthalic anhydride and kept at –20 °C as a 1 N solution in dry THF or toluene.³⁵ *N,N'*-Di-Cbz-*S*-methylisothiurea was synthesized as reported by Tian et al.³² Anhydrous reactions were performed under an argon atmosphere; glassware was flame-dried prior to use under a stream of nitrogen.

(2*S*,3*R*)-3-Nitropropyl-1-(1-phenylethyl)pyrrolidine-2-carboxylic Acid Benzyl Ester (4). To a solution of **1a** (3.22 g, 10 mmol) in dry THF (20 mL) at –78 °C was added LDA (5 mL, 2 N in heptane). A solution of ZnBr₂ (6.76 g, 3 equiv) in dry Et₂O (30 mL) was then added, and the mixture was allowed to warm to rt for 4 h. A solution of CuCN (1.15 g, 1.3 equiv) and LiCl (1.15 g, 2.6 equiv) in dry THF (10 mL) was added at 0 °C. The mixture was cooled to –78 °C, and a

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solution of nitroethylene (10 mL, 1 N in dry THF) was then added. After 15 min of stirring at -78°C , the mixture was quenched with saturated aqueous NH_4Cl . Et_2O was added, and the organic layer was washed with $\text{NH}_4\text{Cl}/\text{NH}_4\text{OH}$ (1 M), 2/1, until complete decoloration followed by brine, dried over MgSO_4 , and concentrated in vacuo. The residue was purified by flash chromatography (cyclohexane/ethyl acetate, 87/13) to give **4** as a yellow oil (1.19 g, 30%) and **5**^{16a} (1.26 g, 39%). Data for **4**: $[\alpha]_{20}^{\text{D}} -67.8$ (c 1, CHCl_3); ^1H NMR (250 MHz, CDCl_3) δ 7.37–7.15, (m, 10H), 5.12–4.96 (AB, 2H, $^3J = 12.5$ Hz), 4.08 (t, 2H, $^3J = 6.25$ Hz), 3.69 (q, 1H, $^3J = 6.5$ Hz), 3.39 (d, 1H, $^3J = 7.5$ Hz), 3.12–2.94 (m, 2H), 2.48–2.18 (m, 1H), 2.10–1.91 (m, 1H), 1.92–1.75 (m, 2H), 1.76–1.52 (m, 1H), 1.33 (d, 3H, $^3J = 6.5$ Hz), 1.32–1.12 (m, 1H); ^{13}C NMR (62.5 MHz, CDCl_3) δ 172.6, 144.4, 135.7, 129.0, 128.5, 128.4, 128.3, 127.3, 127.1, 75.2, 66.1, 65.8, 61.3, 49.8, 41.9, 29.6, 27.7, 26.0, 22.9. Anal. Calcd for $\text{C}_{23}\text{H}_{28}\text{N}_2\text{O}_4$: C, 69.68; H, 7.12; N, 7.07. Found: C, 69.52; H, 7.32; N, 6.88.

Reversibility of the Michael Addition of the Copper–Zinc Reagent to Nitroethylene. The same protocol as for compound **4** was followed. Fifteen minutes after the addition of nitroethylene, TLC indicated complete formation of **4**. TsCN (1 equiv) was added, and the mixture was stirred at -78°C for 2 h. After the reaction was quenched with saturated aqueous NH_4Cl and workup, the residue was purified by flash chromatography (cyclohexane/ethyl acetate, 87/13). Only 2% of the nitro adduct **4** was isolated. A 14% yield of **5** was isolated as well as 38% of **6**, the product resulting from the reaction of the copper–zinc reagent with TsCN .^{16c}

(2S,3R)-3-(N-tert-Butyloxycarbonyl)aminopropyl-1-(1-phenylethyl)pyrrolidine-2-carboxylic Acid Benzyl Ester (7). Zinc powder (5.21 g) was added portionwise at 0°C to a stirred solution of **4** (792 mg, 2 mmol) in AcOH (33 mL). After being stirred overnight at room temperature, the mixture was filtered on a Celite pad, which was then washed with water (3×20 mL). A solution of K_2CO_3 in water ($1 \text{ g}\cdot\text{mL}^{-1}$) was then added dropwise at 0°C until pH 8. Dioxane (120 mL) was then added followed by Boc_2O (436 mg, 2 mmol). After the resulting mixture was stirred for 2 h at room temperature, the organic layer was separated and the aqueous layer was extracted with AcOEt . The combined organic layers were washed with brine, dried over MgSO_4 , and concentrated in vacuo. The crude product was purified by flash chromatography (cyclohexane/ethyl acetate, 84/16) to give **7** as a colorless oil (831 mg, 89%): $[\alpha]_{20}^{\text{D}} -44.9$ (c 1, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ 7.37–7.17, (m, 10H), 5.09–5.01 (AB, 2H), 4.33 + 1.74 (2 br s, NH , 1H, Boc *cis-trans* isomerization), 3.68 (q, 1H, $^3J = 6.5$ Hz), 3.39 (d, 1H, $^3J = 8.1$ Hz), 3.08–2.90 (m, 4H), 2.33–2.22 (m, 1H), 2.04–1.96 (m, 1H), 1.67–1.57 (m, 1H), 1.41 (s, 9H), 1.49–1.17 (m, 3H), 1.33 (d, 3H, $^3J = 6.5$ Hz), 1.09–1.02 (m, 1H); ^{13}C NMR (100 MHz, CDCl_3) δ 173.1, 155.8, 144.6, 135.9, 128.9, 128.5, 128.3, 128.2, 127.4, 127.1, 79.0, 66.5, 65.7, 61.5, 50.1, 42.2, 40.4, 29.8, 28.7, 28.4, 28.2, 22.9. Anal. Calcd for $\text{C}_{23}\text{H}_{28}\text{N}_2\text{O}_4$: C, 72.07; H, 8.21; N, 6.00. Found: C, 72.22; H, 8.05; N, 6.00.

(2S,3R)-3-(N-tert-Butyloxycarbonyl)aminopropyl-1-(9-fluorenylmethyloxycarbonyl)pyrrolidine-2-carboxylic Acid Benzyl Ester (8). A mixture of **7** (700 mg, 1.5 mmol) and Pd/C (150 mg) in methanol (7.5 mL) was stirred overnight under H_2 . The mixture was filtered on a Celite pad and concentrated under reduced pressure to give a white solid. The product was dissolved in H_2O (24 mL), and the solution was cooled to 0°C . NaHCO_3 (378 mg, 4.5 mmol) was then added, and a solution of $\text{Fmoc-O-succinimide}$ (552 mg, 1.65 mmol) in dioxane (24 mL) was added dropwise over 15 min. After the resulting solution was stirred for 7 h at room temperature, the solvent was evaporated and the residue was taken up in water. The aqueous layer was acidified with 1 N HCl until pH 2 and extracted with AcOEt . The combined organic layers were washed with brine, dried over MgSO_4 , and concentrated under reduced pressure. The crude oil was purified by flash chromatography ($\text{CH}_2\text{Cl}_2/\text{EtOH}/\text{acetic acid}$, 10/0.1/0.1) and

recrystallized from $\text{Et}_2\text{O}/\text{pentane}$ to give **8** as a white solid (498 mg, 67%): mp, degradation over 80°C ; $[\alpha]_{20}^{\text{D}} 7.8$ (c 1, CHCl_3); ^1H NMR (400 MHz, CDCl_3) δ 7.75–7.73 (m, 2H), 7.61–7.52 (m, 2H), 7.41–7.33 (m, 2H), 7.29–7.26 (m, 2H), 6.73 + 6.42 + 4.68 (3 br s, NH , 1H, Boc and Fmoc cis-trans isomerization), 4.47–4.16 (m, 4H), 3.79–3.77 (m, 1H), 3.43–3.39 (m, 1H), 3.21–3.01 (m, 2H), 2.43–2.27 (m, 1H), 2.09–1.96 (m, 1H), 1.94–1.71 (m, 1H), 1.69–1.48 (m, 2H), 1.41 + 1.46 = 1.43 (3s, 9H, Boc and Fmoc cis-trans isomerization), 1.47–1.27 (m, 2H); ^{13}C NMR (100 MHz, CDCl_3) δ 174.7, 174.4, 158.5, 156.1, 154.9, 154.5, 143.9, 141.3, 141.2, 141.2, 127.7, 127.1, 125.2, 125.1, 119.9, 80.9, 80.9, 79.5, 79.3, 67.7, 62.4, 62.1, 47.3, 47.2, 46.4, 46.0, 43.0, 41.6, 40.4, 29.8, 28.6, 28.4, 28.1, 27.5, 27.3, 27.1; ESI^+ MS m/z 517.16 $[\text{MNa}]^+$. Anal. Calcd for $\text{C}_{28}\text{H}_{34}\text{N}_2\text{O}_6$: $0.5\text{H}_2\text{O}$: C, 66.85; H, 6.96; N, 5.57. Found: C, 66.98; H, 7.27; N, 5.45.

(2S,3S)-3-Carboxymethyl-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid (9). A mixture of **6**^{16c} (1.046 g, 3 mmol) and 6 N HCl (30 mL) were refluxed for 3 h. The solvent was evaporated and the crude product dissolved in MeOH (15 mL). K_2CO_3 (830 mg, 6 mmol) and Boc_2O (657 mg, 3 mmol) were added, and the mixture was stirred overnight under H_2 . The mixture was filtered on a Celite pad and concentrated in vacuo. The crude product was taken up in water and washed with Et_2O . The aqueous layer was acidified at 0°C with 1 N HCl and extracted with CH_2Cl_2 (many extractions were required), and the combined organic layers were dried over MgSO_4 and concentrated in vacuo. The residue was crystallized from Et_2O to give **9** as a white solid (383 mg, 47%): mp 152–155 $^{\circ}\text{C}$; $[\alpha]_{20}^{\text{D}} 82$ (c 1, MeOH); ^1H NMR (250 MHz, CDCl_3) δ 4.51 and 4.38 (2d, Boc *cis-trans* isomerization, 1H, $^3J = 8.1$ and 8.1 Hz), 3.71–3.59 (m, 1H), 3.42–3.30 (m, 1H), 2.97–2.79 (m, 1H), 2.64–2.46 (m, 2H), 2.10–2.04 (m, 1H), 1.86–1.73 (m, 1H), 1.46–1.42 (2s, Boc *cis-trans* isomerization, 9H); ^{13}C NMR (62.5 MHz, CDCl_3) δ 176.5, 176.4, 175.2, 155.0, 154.0, 80.9, 80.7, 61.6, 61.1, 45.8, 45.4, 38.0, 37.3, 34.6, 29.7, 28.9, 28.4, 28.3. Anal. Calcd for $\text{C}_{12}\text{H}_{19}\text{NO}_6$: C, 52.74; H, 7.01; N, 5.13. Found: C, 52.45; H, 7.19; N, 5.04.

(2S,3S)-3-Benzoyloxycarbonylmethyl-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid Benzyl Ester (10). To a solution of **9** (273 mg, 1 mmol) in MeOH (1 mL) and H_2O (0.1 mL) was added a 20% aqueous solution of CsCO_3 until pH 7. The solvent was removed in vacuo and the mixture taken in DMF (2 mL). BnBr (241 μL , 2.2 mmol) was then added, and the mixture was stirred at rt overnight. Et_2O was added, and the organic layer was washed with saturated aqueous NH_4Cl . The crude oil was purified by flash chromatography (cyclohexane/ethyl acetate, 85/15) to give **10** as a colorless oil (340 mg, 75%): $[\alpha]_{25}^{\text{D}} 21.3$ (c 1, CHCl_3); ^1H NMR (250 MHz, CDCl_3) δ 7.40–7.25 (m, 10H), 5.22–4.96 (m, 4H), 4.47 and 4.39 (2d, Boc *cis-trans* isomerization, 1H, $^3J = 8.3$ and 8.3 Hz), 3.76–3.58 (m, 1H), 3.41–3.25 (m, 1H), 2.94–2.78 (m, 1H), 2.49–2.08 (m, 2H), 2.08–1.56 (m, 2H), 1.45–1.32 (2s, Boc *cis-trans* isomerization, 9H); ^{13}C NMR (62.5 MHz, CDCl_3) δ 171.6, 171.3, 154.4, 153.8, 135.8, 135.5, 128.8, 128.7, 128.6, 128.5, 128.4, 80.2, 80.0, 66.8, 66.6, 61.9, 61.6, 45.9, 45.5, 38.6, 37.8, 35.0, 29.9, 29.0, 28.5, 28.3. Anal. Calcd for $\text{C}_{26}\text{H}_{31}\text{NO}_6$: C, 68.86; H, 6.89; N, 3.09. Found: C, 68.89; H, 7.05; N, 3.10.

(2S,3S)-3-Carboxymethyl-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid Benzyl Ester (11). To a solution of **10** (272 mg, 0.6 mmol) in acetone/water (6 mL, 4/1) was slowly added 56% LiOH (25 mg, 0.6 mmol) in water (1.2 mL). After the resulting solution was stirred overnight at rt, the solvent was evaporated in vacuo and water was added. The aqueous layer was washed with Et_2O , acidified at 0°C with 1 N HCl until pH 2, and extracted with AcOEt . The combined AcOEt layers were dried over MgSO_4 and concentrated in vacuo. The crude oil was purified by flash chromatography ($\text{CH}_2\text{Cl}_2/\text{EtOH}/\text{acetic acid}$, 10/0.1/0.1) to give **11** as a colorless oil (148 mg, 68%): $[\alpha]_{25}^{\text{D}} 13.5$ (c 1, CHCl_3); ^1H NMR (250 MHz, CDCl_3) δ 7.37–7.27 (m, 5H), 5.29–5.05 (m, 2H), 4.50 and 4.39 (2d, Boc *cis-trans* isomerization, 1H, $^3J = 8.3$ and

8.0 Hz), 3.76–3.60 (m, 1H), 3.40–3.26 (m, 1H), 2.86–2.75 (m, 1H), 2.42–2.17 (m, 2H), 2.08–1.99 (m, 1H), 1.77–1.66 (m, 1H), 1.46–1.33 (2s, Boc *cis-trans* isomerization, 9H); ^{13}C NMR (62.5 MHz, CDCl_3) δ 176.6, 171.4, 154.4, 153.8, 135.4, 135.2, 128.8, 128.6, 128.5, 128.3, 80.3, 80.2, 66.9, 61.7, 61.4, 45.8, 45.5, 38.3, 37.5, 34.6, 29.7, 28.8, 28.4, 28.2. Anal. Calcd for $\text{C}_{19}\text{H}_{25}\text{NO}_6$: C, 62.80; H, 6.93; N, 3.85. Found: C, 62.76; H, 6.97; N, 3.76.

(2S,3S)-3-Methyloxycarbonylmethyl-1-(1-phenylethyl)-pyrrolidine-2-carboxylic Acid Benzyl Ester (12). Acetyl chloride (7.42 mL, 30 equiv) was added dropwise to a solution of **6**^{16c} (1.20 g, 3.48 mmol) in 17 mL of anhydrous MeOH at 0 °C. The reaction mixture was stirred at 0 °C for 1 h and then for 5 h at rt. The solvent was evaporated in vacuo and the crude product taken up in water. A 10% aqueous solution of K_2CO_3 was added until pH 8. After extraction with Et_2O , the organic layer was washed with brine and dried over MgSO_4 , and the solvent was evaporated in vacuo. The crude oil was purified by flash chromatography (cyclohexane/ethyl acetate, 9/1) to give **12** as a colorless oil (923 mg, 70%): $[\alpha]_{25}^{\text{D}}$ –44.9 (c 1, CHCl_3); ^1H NMR (250 MHz, CDCl_3) δ 7.35–7.15 (m, 10H), 5.12–4.94 (AB, 2H, $^3J = 11.25$ Hz), 3.71 (q, 1H, $^3J = 7.5$ Hz), 3.51 (d, 1H, $^3J = 10$ Hz), 3.53 (s, 3H), 3.12–2.70 (m, 3H), 2.80–2.60 (m, 1H), 2.37–2.00 (m, 3H), 1.32 (d, 3H, $^3J = 7.5$ Hz); ^{13}C NMR (62.5 MHz, CDCl_3) δ 172.9, 172.5, 144.4, 135.7, 128.6, 128.5, 128.3, 127.3, 65.9, 65.7, 61.4, 51.5, 49.9, 38.1, 35.5, 29.7, 22.8. Anal. Calcd for $\text{C}_{23}\text{H}_{27}\text{NO}_4$: C, 72.42; H, 7.13; N, 3.67. Found: C, 72.28; H, 7.31; N, 3.51.

(2S,3S)-3-Methyloxycarbonylmethyl-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid (13). A mixture of compound **12** (627 mg, 1.64 mmol), 10% Pd/C (164 mg), and Boc_2O (1.5 equiv, 378 mg) in MeOH (8 mL) was stirred overnight at room temperature under hydrogen. After filtration over a Celite pad and evaporation of the solvent in vacuo, the crude product was taken up in water and a 10% aqueous solution of K_2CO_3 was added until pH 8. The aqueous layer was washed with Et_2O and then acidified at 0 °C with 1 N HCl until pH 2. The aqueous layer was extracted with AcOEt, and the combined organic layers were dried over MgSO_4 and concentrated in vacuo to give **13** as a colorless oil. White crystals were obtained after crystallization from Et_2O /pentane (376 mg, 80%): mp 87–90 °C; $[\alpha]_{25}^{\text{D}}$ 25.0 (c 1, CHCl_3); ^1H NMR (250 MHz, CDCl_3) δ 4.44 and 4.37 (2d, Boc *cis-trans* isomerization, 1H, $^3J = 7.5$ and 10 Hz), 3.74–3.58 (m, 1H), 3.70 (s, 3H), 3.38–3.30 (m, 1H), 2.99–2.73 (m, 1H), 2.72–2.28 (m, 2H), 2.16–2.03 (m, 1H), 1.88–1.64 (m, 1H), 1.46–1.42 (2s, Boc *cis-trans* isomerization, 9H); ^{13}C NMR (62.5 MHz, CDCl_3) δ 176.6, 175.3, 172.1, 154.8, 153.8, 80.6, 61.6, 61.2, 51.9, 45.9, 45.5, 38.3, 37.4, 34.6, 29.7, 28.9, 28.4, 28.3; DCI MS m/z 288 $[\text{MH}^+]$. Anal. Calcd for $\text{C}_{13}\text{H}_{21}\text{NO}_6 \cdot \text{H}_2\text{O}$: C, 51.43; H, 7.58; N, 4.61. Found: C, 51.70; H, 7.30; N, 4.81.

(2S,3S)-3-Cyanomethyl-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid (14). **6**^{16c} (4 g, 11.6 mmol), 10% Pd/C (1.16 g), Boc_2O (2.54 g, 11.6 mmol), and 55 mL of MeOH were stirred overnight under H_2 . The reaction mixture was then filtered on a Celite pad, and the solvent was evaporated in vacuo. The crude product was taken up in water, and a 10% aqueous solution of K_2CO_3 was added until pH 8. The aqueous layer was washed with Et_2O , acidified at 0 °C with 1 N HCl until pH 2, and extracted with AcOEt. The combined AcOEt layers were washed with brine and dried over MgSO_4 , and the solvent was evaporated in vacuo. The crude product was recrystallized from Et_2O to give **14** as a white solid (2.21 g, 75%): mp 162 °C; $[\alpha]_{25}^{\text{D}}$ 2.3 (c 1, CHCl_3); ^1H NMR (250 MHz, CDCl_3) δ 4.41 and 4.35 (2d, Boc *cis-trans* isomerization, 1H, $^3J = 7.5$ and 8.0 Hz), 3.80–3.58 (m, 1H), 3.52–3.32 (m, 1H), 2.87–2.56 (m, 2H), 2.56–2.33 (m, 1H), 2.31–2.08 (m, 1H), 2.06–1.80 (m, 1H), 1.46–1.42 (2s, Boc *cis-trans* isomerization, 9H); ^{13}C NMR (62.5 MHz, CDCl_3) δ 175.2, 174.0, 154.9, 153.9, 117.8, 117.6, 81.4, 81.2, 61.5, 61.1, 45.8, 45.4, 38.6, 37.8, 29.6, 28.8, 28.5, 28.4, 18.4. Anal. Calcd for $\text{C}_{12}\text{H}_{18}\text{N}_2\text{O}_4$: C, 56.68; H, 7.13; N, 11.02. Found: C, 56.83; H, 7.38; N, 10.83.

(2S,3S)-3-Acetamide-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid (15). To a solution of **14** (509 mg, 2 mmol) in 0.5 N NaOH (6 mL) was added 35% H_2O_2 (6 mL) followed by 2 N NaOH (4 mL). After being stirred for 30 min at rt, the mixture was cooled to 0 °C and 1 N HCl was added until pH 2. The aqueous layer was extracted with CH_2Cl_2 (many extractions were required). The combined organic layers were dried on MgSO_4 and concentrated in vacuo. The crude product was recrystallized with Et_2O /pentane to give **15** as a white solid (327 mg, 60%): mp, degradation over 110 °C; $[\alpha]_{25}^{\text{D}}$ 21.5 (c 1, CHCl_3); ^1H NMR (500 MHz, CDCl_3) δ 9.71 (br s, 1H), 7.02 (d, 1H, $^3J = 21.0$ Hz), 6.75 (d, 1H, $^3J = 19.7$ Hz), 4.38 and 4.33 (2d, Boc *cis-trans* isomerization, 1H, $^3J = 8.1$ and 8.4 Hz), 3.68–3.58 (m, 1H), 3.36–3.29 (m, 1H), 2.89–2.82 (m, 1H), 2.49–2.35 (m, 2H), 2.05–2.00 (m, 1H), 1.81–1.71 (m, 1H), 1.44 and 1.40 (2s, Boc *cis-trans* isomerization, 9H); ^{13}C NMR (100 MHz, CDCl_3) δ 175.8, 174.8, 154.9, 154.1, 80.6, 80.4, 62.0, 61.6, 45.8, 45.4, 38.6, 37.8, 35.8, 35.7, 29.8, 29.0, 28.5, 28.4; ESI⁺ MS m/z 271.06 $[\text{M} - \text{H}]^+$.

(2S,3R)-3-Aminoethyl-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid (16). A mixture of **14** (509 mg, 2 mmol), PtO_2 (200 mg), and MeOH (10 mL) was stirred overnight under 5 bar of H_2 . The mixture was filtered on a Celite pad, and the solvent was evaporated in vacuo. The resulting white solid was washed with Et_2O to give **16** (496 mg, 96%): mp, degradation over 200 °C; $[\alpha]_{25}^{\text{D}}$ 26 (c 0.3, H_2O); ^1H NMR (250 MHz, D_2O) δ 4.11 and 4.10 (2d, Boc *cis-trans* isomerization, 1H, $^3J = 8.5$ and 8.5 Hz), 3.70–3.50 (m, 1H), 3.39–3.16 (m, 1H), 3.08 (t, 2H, $^3J = 8$ Hz), 2.52–2.32 (m, 1H), 2.08–1.94 (m, 1H), 1.91–1.49 (m, 3H), 1.43–1.39 (2s, Boc *cis-trans* isomerization, 9H); ^{13}C NMR (62.5 MHz, D_2O) δ 178.6, 156.4, 81.8, 65.1, 64.6, 46.3, 45.7, 45.5, 39.6, 38.7, 38.4, 29.6, 29.1, 28.4, 28.2, 28.0. Anal. Calcd for $\text{C}_{13}\text{H}_{21}\text{NO}_6 \cdot \text{H}_2\text{O}$: C, 52.1; H, 8.60; N, 10.10. Found: C, 52.34; H, 8.31; N, 9.96.

(2S,3R)-3-*N,N'*-Dicarboxybenzylguanidinoethyl-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid (17). To a solution of **16** (259 mg, 1 mmol) in dry THF (5 mL) under argon were added *N,N'*-di-Cbz-*S*-methylisothiourea³² (388 mg, 1.1 mmol) and K_2CO_3 (276 mg, 2 mmol). The reaction mixture was stirred at rt for 6 days. The solvent was removed under reduced pressure, and the crude mixture was partitioned between water and Et_2O . The aqueous layer was washed with Et_2O , acidified at 0 °C with 1 N HCl, and extracted with EtOAc. The combined AcOEt layers were dried over MgSO_4 and concentrated in vacuo. The residue was crystallized from Et_2O /pentane to give **17** as a white solid (361 mg, 61%): mp 93–94 °C; $[\alpha]_{25}^{\text{D}}$ 24.8 (c 1, CHCl_3); ^1H NMR (250 MHz, CDCl_3) 11.72 (br s, 1H), 8.41–8.29 (m, 2H), 7.39–7.27 (m, 10H), 5.17 (s, 2H), 5.12 (s, 2H), 4.36 and 4.28 (2d, Boc *cis-trans* isomerization, 1H, $^3J = 8.4$ and 8.3 Hz), 3.76–3.42 (m, 3H), 3.33–3.24 (m, 1H), 2.47–2.34 (m, 1H), 2.05–1.99 (m, 1H), 1.87–1.73 (m, 2H), 1.61–1.44 (m, 1H), 1.45–1.40 (2s, Boc *cis-trans* isomerization, 9H); ^{13}C NMR (62.5 MHz, CDCl_3) δ 176.2, 175.0, 163.5, 156.1, 154.7, 153.9, 153.7, 136.6, 134.5, 128.8, 128.7, 128.4, 128.1, 128.0, 80.4, 68.3, 67.2, 62.1, 61.6, 46.0, 45.6, 40.0, 39.5, 39.2, 29.7, 29.5, 29.1, 28.4, 28.3. Anal. Calcd for $\text{C}_{28}\text{H}_{35}\text{N}_4\text{O}_8$: C, 60.53; H, 6.35; N, 10.08. Found: C, 60.81; H, 6.23; N, 9.77.

(2S,3R)-2-Oxotetrahydrofuran(2,3)-1-(1-phenylethyl)-pyrrolidine (18) and (2S,3R)-3-Hydroxymethyl-1-(1-phenylethyl)pyrrolidine-2-carboxylic Acid Methyl Ester (19). To a solution of **1b** (1.61 g, 5 mmol) in dry THF (10 mL) at –78 °C was added LDA (2.5 mL, 2 N in heptane). A solution of ZnBr_2 (3.38 g, 3 equiv) in dry Et_2O (15 mL) was then added, and the mixture was allowed to warm to rt for 4 h. The solution was cooled to 0 °C, and dioxygen was bubbled for 30 min. The reaction was then quenched with saturated aqueous NH_4Cl . Et_2O was added, and the organic layer was washed with saturated aqueous NH_4Cl , dried over MgSO_4 , and concentrated under reduced pressure. The residue was purified by flash chromatography (cyclohexane/ethyl acetate, 85/15) to give a 6/4 mixture of **18** and **19** (731 mg, 60%). **18** was isolated for

analysis as a pale yellow solid: mp 65 °C; $[\alpha]_{20}^D$ -62.9 (c 1, CHCl₃); ¹H NMR (250 MHz, CDCl₃) δ 7.42–7.24, (m, 5H), 4.31 (t, 1H, ³J = 9.5 Hz), 4.17 (q, 1H, ³J = 7.5 Hz), 4.11 (dd, 1H, ³J = 2.5, 9.5 Hz), 3.50 (d, 1H, ³J = 7.5 Hz), 2.95–2.83 (m, 1H), 2.82–2.75 (m, 2H), 2.16–2.08 (m, 1H), 1.75–1.70 (m, 1H), 1.46 (d, 3H, ³J = 7.5 Hz); ¹³C NMR (62.5 MHz, CDCl₃) δ 176.7, 143.0, 128.2, 127.9, 127.1, 72.0, 61.8, 59.4, 50.0, 38.7, 30.87, 22.8. Anal. Calcd for C₂₃H₂₈N₂O₄: C, 72.73; H, 7.36; N, 6.06. Found: C, 72.73; H, 7.48; N, 5.94. Data for **19**: ¹H NMR (250 MHz, CDCl₃) δ 7.41–7.20, (m, 5H), 4.01 (q, 1H, ³J = 7.5 Hz), 3.94 (d, 1H, ³J = 5 Hz), 3.70 (s, 3H), 3.34 (d, 1H, ³J = 7.5 Hz), 3.20–2.83 (m, 1H), 2.83–2.53 (m, 3H), 1.99–1.56 (m, 2H), 1.43 (d, 3H, ³J = 7.5 Hz).

(2S,3R)-2-Oxotetrahydrofuran(2,3)-1-(tert-butyloxycarbonyl)pyrrolidine (20) and (2S,3R)-3-Hydroxymethyl-1-(tert-butyloxycarbonyl)pyrrolidine-2-carboxylic Acid Methyl Ester (21). A mixture of **18** + **19** (610 mg, 2.5 mmol) and Pd/C (250 mg) in methanol (12.5 mL) was stirred overnight under H₂. The mixture was filtered on a Celite pad and concentrated under reduced pressure to give a white oil. The residue was purified by flash chromatography (cyclohexane/ethyl acetate/acetic acid, 6/4/0.1) to give a 1/1 mixture of **20** and **21** (547 mg, 90%). **20** was further isolated for analysis as a white solid: mp 93 °C; $[\alpha]_{20}^D$ -114.1 (c 1, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ 4.78/4.52 (m, 1H), 4.43 (t, 1H, ³J = 9.5 Hz), 4.14 (dd, 1H, ³J = 2.5, 9.5 Hz), 3.60–3.45 (m, 2H), 3.22–3.07 (m, 1H), 2.31–2.19 (m, 1H), 1.92–1.78 (m, 1H), 1.49–1.41 (2s, Boc *cis-trans* isomerization, 9H); ¹³C NMR (100 MHz, CDCl₃) δ 174.0, 154.1, 154.0, 80.6, 69.4, 59.0, 46.8, 46.0, 39.2, 38.3, 29.9, 29.8, 28.8, 28.3. Anal. Calcd for C₂₃H₂₈N₂O₄: C, 58.14; H, 7.54; N, 6.16. Found: C, 58.12; H, 7.59; N, 6.12. Data for **21**: ¹H NMR (250 MHz, CDCl₃) δ 4.47–4.37 (m, 2H), 3.75 (s, 3H), 3.75–3.28 (m, 3H), 2.85–2.61 (m, 2H), 2.07–1.60 (m, 2H), 1.46–1.41 (2s, Boc *cis-trans* isomerization, 9H).

(2S,3R)-3-Benzoyloxymethyl-1-(tert-butyloxycarbonyl)-pyrrolidine-2-carboxylic Acid Benzyl Ester (22). To a mixture of **20** + **21** (486 mg, 2 mmol), benzyl bromide (0.956 mL, 8 mmol), and anhydrous toluene (8 mL) was added freshly crushed KOH (476 mg, 8.5 mmol). The mixture was heated at

80 °C for 8 h. After being cooled to room temperature, the mixture was partitioned between Et₂O and water. The aqueous layer was extracted with Et₂O, and the combined organic layers were washed with brine, dried over MgSO₄, and concentrated under reduced pressure. The residue was purified by flash chromatography (cyclohexane/ethyl acetate, 85/15) to give **22** (672 mg, 79%): $[\alpha]_{20}^D$ 1.1 (c 1, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ 7.32–7.22, (m, 10H), 5.18–4.97 (m, 2H), 4.44–4.29 (m, 3H), 3.73–3.61 (m, 1H), 3.46–3.24 (m, 3H), 2.78–2.63 (m, 1H), 2.07–1.82 (m, 2H), 1.45–1.34 (2s, Boc *cis-trans* isomerization, 9H); ¹³C NMR (100 MHz, CDCl₃) δ 171.6, 171.4, 154.3, 153.8, 138.0, 137.9, 135.8, 135.6, 128.7, 128.5, 128.4, 128.4, 128.1, 127.7, 127.6, 127.5, 80.0, 79.9, 73.2, 73.1, 69.2, 69.1, 66.6, 61.1, 60.6, 45.9, 45.5, 43.2, 42.2, 28.4, 28.2, 27.6, 26.6. Anal. Calcd for C₂₅H₃₁NO₄: C, 70.57; H, 7.34; N, 3.29. Found: C, 70.20; H, 7.67; N, 3.27.

(2S,3R)-3-Benzoyloxymethyl-1-(tert-butyloxycarbonyl)-pyrrolidine-2-carboxylic Acid (23). To a solution of **22** (596 mg, 1.4 mmol) in acetonitrile/water (4/1, 5 mL) was added LiOH·H₂O (294 mg, 7 mmol), and the mixture was refluxed overnight. The solvent was evaporated and the residue taken up in water. After acidification to pH 2 with 1 N HCl, the aqueous layer was extracted with Et₂O. The combined organic layers were washed with brine, dried over MgSO₄, and concentrated in vacuo. The residue was purified by flash chromatography (cyclohexane/ethyl acetate/acetic acid, 8/2/0.1) to give **23** (376 mg, 80%) as a white solid: $[\alpha]_{20}^D$ 14.5 (c 1, CHCl₃); ¹H NMR (400 MHz, CDCl₃) δ 8.87 (br s, 1H), 7.35–7.27, (m, 5H), 4.49–4.43 (m, 2H), 4.42 and 4.34 (2d, Boc *cis-trans* isomerization, 1H, ³J = 8.3 and 8.4 Hz), 3.71–3.33 (m, 4H), 2.83–2.65 (m, 1H), 2.10–1.88 (m, 2H), 1.45–1.41 (2s, Boc *cis-trans* isomerization, 9H); ¹³C NMR (100 MHz, CDCl₃) δ 177.0, 175.7, 154.9, 154.0, 138.1, 138.0, 128.6, 127.9, 127.8, 80.6, 73.5, 69.6, 69.4, 61.1, 60.6, 46.1, 45.7, 43.1, 42.1, 28.6, 28.5, 27.8, 26.8. Anal. Calcd for C₁₈H₂₅NO₅: C, 64.46; H, 7.51; N, 4.18. Found: C, 64.08; H, 7.62; N, 4.03.

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